AI TOOLS FOR RESEARCH

PROMPT 06
DATASET & BASELINE VALIDATION

Diagnosing Capability and Specification Uncertainty in Tool Retrieval: An Evidence-Grounded Evaluation

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Verdict: CONDITIONAL GO for source-derived annotation feasibility only. No validated three-action dataset or executable baseline has yet been established.

Scope: Audit of sources, repository files, label ontology, candidate comparators, licensing and split design. No Prompt 07 protocol freeze, model training or performance measurements.

1 Question, evidence standard and decision

Primary question: Does explicit capability/requirement evidence improve SELECT / NO_VALID_TOOL / NEEDS_CLARIFICATION decisions over development-calibrated retrieval-confidence policies?

VERIFIED: Actual source-file JSON samples and original scholarly pages inspected. **INFERRED:** viable methods and compute estimates. **UNKNOWN:** annotation feasibility, licensing beyond code, full splits, baseline runtime, model weights and novelty.

2 Dataset comparison and verification matrix

Dataset	Verified source/protocol	Current label coverage	Ontology suitability / verdict
MetaTool / ToolE	Public ICLR 2024 repo; tool labels/descriptions; 199 descriptions, 497 multi-tool records parsed. Single-tool CSV not parsed.	Tool identity for original queries; four tool-choice subtasks. Negative capability / clarification not confirmed.	Candidate source for manual annotation; CONDITIONAL only. Requires ID joins and independently supported negatives.
ToolRet	ACL 2025 repo + documented example with query, positive relevance, tool schema/parameters; released files not parsed.	Retrieval relevance rather than three decision actions; multi-positive completeness uncertain.	Useful parameter-rich secondary source and retrieval baseline reference; three-class suitability UNKNOWN.
ToolEX / ToolEQ	2026 preprint abstract: equivalent-tool reference expansion; no dataset inspected.	Addresses alternative positive references; no proof of three-class action labels.	Potential multi-valid validation source; access, license, ontology fit UNKNOWN.
AgentAbstain	2026 preprint: 263 paired agent tasks/42 sandbox environments, author-reported.	Act/abstain-oriented paired tasks; full inventory/capability labels not checked.	Relevant comparator concept; heavier execution and mismatch with retrieval-stage actions.
ClarifyBench / SAGE-Agent	ACL 2026 published parameter-uncertainty clarification study; source page checked.	Underspecification/clarification focus; exact dataset fields not inspected.	Closest clarification data candidate; capability-absence label and redistribution UNKNOWN.

3 Direct file-level findings

MetaTool `plugin_des.json`: parsed JSON object: 199 entries, no JSON null values; key `timeport` describes a time-travel game, while `airqualityforeast` describes a two-day US ZIP-code air-quality forecast. SHA 0c086d1a5e7b1586e8d5cafdf044871fa07aa8b8.

MetaTool `multi_tool_query_golden.json`: parsed 497 array records. First genuine record asks about recent Tesla news and stock impact, with original labels `FinanceTool`, `NewsTool`. SHA 193d1f6ef9ab52d6bc7e51100c69c0d9ab2de259.

MetaTool `plugin_info.json`: parsed 390 records, including model-facing description fields. Stable joins with the 199 tool IDs not tested. SHA 20cbad749a67a3c2ecbdc6a9e2f6342a92e055a0.

Single-tool CSV: repository lists `dataset/data/all_clean_data.csv` (SHA ae09901a3522df5ed0f83655f57267a4d6775e16), but GitHub connector yielded an empty content payload. The earlier historical Colab observation of 20,614 rows versus README 20,630 remains unresolved.

ToolRet: README publicly includes example `apigen_query_5` (bacterial growth, parameter-rich schema) and `intfloat/e5-small-v2` evaluation example; actual full data and weights were not parsed or run.

4 Feasibility of action labels

Action	Required evidence	Critical failure mode
SELECT	At least one available tool demonstrably fits all mandatory requirements; enough information to choose. Preserve all supported positives.	Original single-gold ID may omit valid alternatives.
NO_VALID_TOOL	Complete candidate inventory examined; documented conflicts or absence of mandatory capability established.	Missing description is not proof of impossibility.
NEEDS_CLARIFICATION	Missing user requirement changes justified selection among plausible available options; document the missing slot.	Low confidence / multiple valid tools alone does not imply asking.
MULTI_VALID	Property of SELECT: more than one supported valid tool, not a distinct action.	One-to-one scoring rejects genuinely correct alternatives.
INSUFFICIENT_EVIDENCE	Tool documentation is too incomplete to make a justified label; withhold from primary evaluation.	Invented negative labels cause non-reproducible conclusions.

Annotation feasibility: No human-validated labels, class counts, inter-annotator scores, or pilot outcomes exist. Proposed 30–50-family exploratory pilot with independent reviews, documented evidence, adjudication, exclusion accounting and agreement measurement. Detailed schema: accompanying `P06_Annotation_Rubric.md`.

5 Comparator validation

Comparator	Use and fairness	Access/execution
Forced BM25 or dense top-1	Diagnostic retrieval reference; cannot by itself output all three actions.	Implementable in principle; not executed.
Max-score / margin policy	Simple selective baseline; three-way mapping must be explicit and testable, not invented post hoc.	Proposed, not run; score scales model dependent.
Development-calibrated three-way policy	PRIMARY COMPARABLE BASELINE. Same retriever; scores, margins and inventory features; development-only calibration.	No labels yet; not trained or validated.
Evidence-grounded diagnostic	Proposed method: documented capability/parameter/requirement features; ablate features later.	Not implemented.
Published stronger controls	Tool-REX, MFTR, SAGE-Agent or AgentAbstain only with aligned action space and input budget.	Task matching, code/weights, compute and rights unverified.

6 Leakage-safe evaluation design

Group all original questions, paraphrases and counterfactual variants into the same partition. Design labels and exclusion rules without consulting held-out model predictions. Fit models and calibrate thresholds on development groups only; leave untouched test families for final evaluation. Consider tool-family or inventory-held-out test as additional generalization measure. No fixed train/dev/test percentages are justified until genuine family counts and grouping structure are known.

Analyze macro-F1 on the three actions, false-selection rate on unsupported requests, inappropriate clarification, paired action-transition consistency, multi-positive-aware Recall@k, and optional risk–coverage and latency. Use family-level paired uncertainty intervals, logged sampling seeds, and avoid treating repeated variants as independent records.

7 Access, licensing, compute and human cost

Item	Known fact	Estimate / unresolved requirement
MetaTool source	Repository marked MIT for code; content JSON inspected.	Dataset provenance / plugin redistribution rights UNKNOWN.
ToolRet source	Repository marked Apache-2.0; sample + environment documented.	Underlying dataset terms may differ; actual weight permissions UNKNOWN.
Embedding weights	`intfloat/e5-small-v2` referenced in README.	Download, revision, memory requirement and license not checked.
Google Colab	User-selected free environment; no diagnostics run.	Small BM25 / calibration likely CPU-feasible; dense model may need RAM/storage; no GPU guaranteed.
Human validation	No annotation conducted.	30–50-family pilot initially proposed; independent review/adjudication likely dominates elapsed effort.
Full evaluation	No baseline or score executed.	Resource footprint, runtime, class balance, calibration complexity UNKNOWN.

8 Outstanding risks and decision gate

High risks: Under-documented or ambiguous tools; false negatives; incomplete positive labels; counterfactual artifacts; source/target leakage; calibration on test data; category imbalance; synthetic provenance; benchmark contamination; task mismatch for published comparators.

CONDITIONAL GO: MetaTool is the best-documented source for a small evidence-audited annotation attempt, not a verified ready-made three-class dataset. The primary fair control is the development-calibrated three-way policy, supported by a transparent score/margin selective baseline. A forced top-1 selector is supplementary only.

Do not freeze Prompt 07 until: (1) real single-tool records loaded; (2) counts and joins reconciled; (3) dataset and model rights clarified; (4) independent pilot confirms justified three-class labels; (5) fair baseline mappings and runtime are demonstrated; (6) closest prior-art overlap reviewed.

9 Source provenance

MetaTool (Huang et al., 2024): https://github.com/HowieHwong/MetaTool ; https://arxiv.org/abs/2310.03128

ToolRet (Shi et al., 2025): https://github.com/mangopy/tool-retrieval-benchmark ; https://aclanthology.org/2025.findings-acl.1258/

ToolEX preprint: https://arxiv.org/abs/2609.08327 ; AgentAbstain: https://arxiv.org/abs/2607.10059

Clarification (Suri et al., 2026): https://aclanthology.org/2026.findings-acl.2028/

Detailed inspection commands, SHA identifiers, unverified claims and next-stage requirements are preserved in the companion Markdown records.